FRANKENSNARE USER GUIDE



Frankensnare lets you build larger-than-life snares by putting together the most desirable sonic elements of a huge variety of snare drums. It uses samples of acoustic snares the way Dr. Viktor Frankenstein tried to use the best parts of various humans. We have succeeded where he failed, though we both created monsters in the process. Frankensnare requires Plogue Sforzando version 1.960 or higher, which is free and can be downloaded from https://plogue.com/downloads.html

Any questions or comments? Contact us at samples@karoryfer.com

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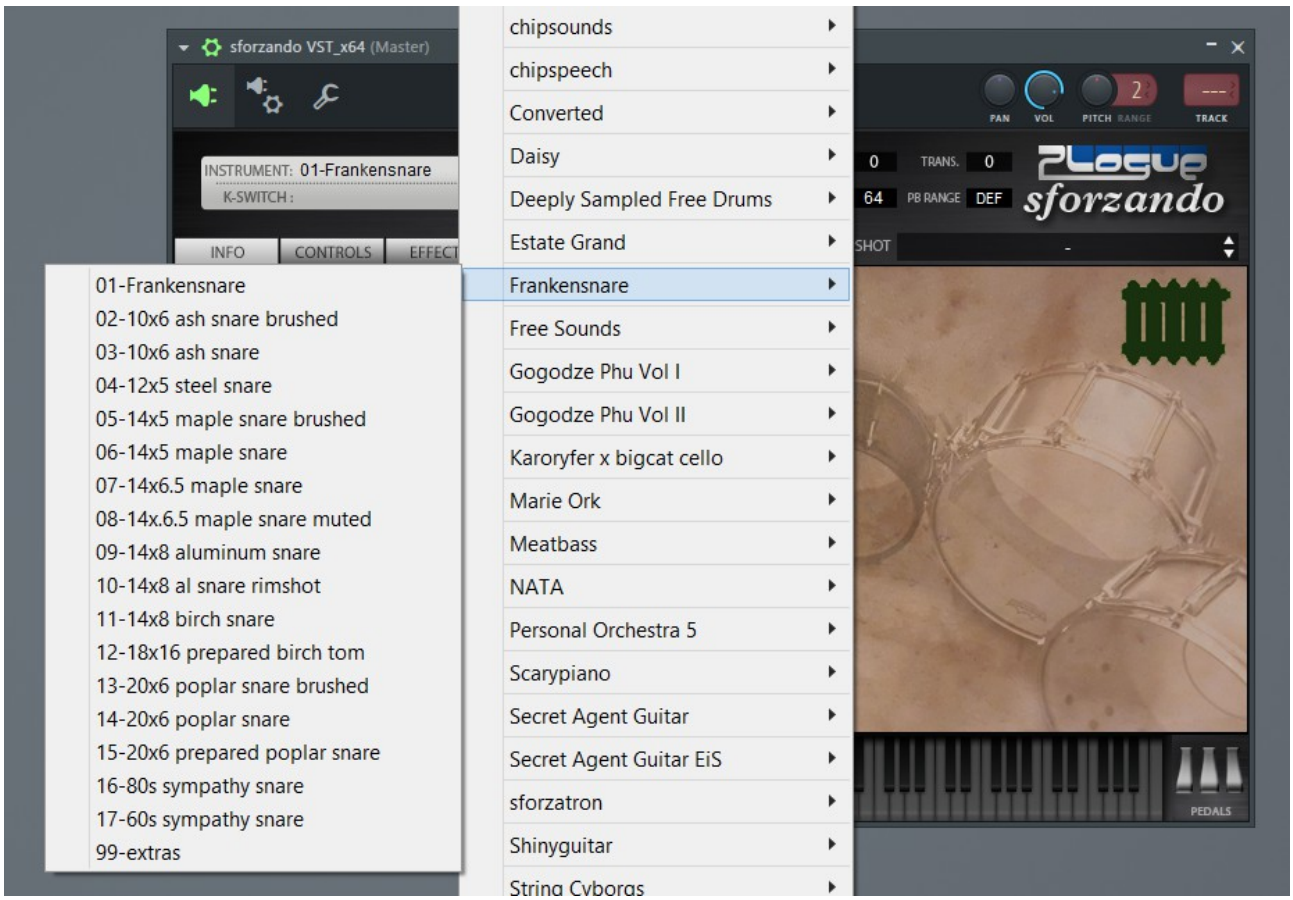
INSTALLATION AND REGISTRATION

If you do not have Sforzando installed, install that first – version 1.960 or newer is required. After downloading and unzipping the Frankensnare content into a folder, launch Sforzando, either in standalone mode or as a plugin. It is important that all the files stay in the same folder – changing the folder structure would require editing the SFZ files. Drag the Frankensnare.bank.xml file onto the Sforzando interface – if this does not work in your DAW, try doing it with Sforzando in standalone mode. After providing administrator credentials, the instrument will be registered. The SFZ files and WAV files can be used without registration, but registration causes Frankensnare to appear in Sforzando's instrument list, which allows the instruments to be loaded with the GUI.

If you are an owner of Frankensnare version 1.000 and are upgrading, you can extract the new version's files into the same folder where version 1.000 was installed. The next time you launch Sforzando, the new instrument bank and all its contents should be available.

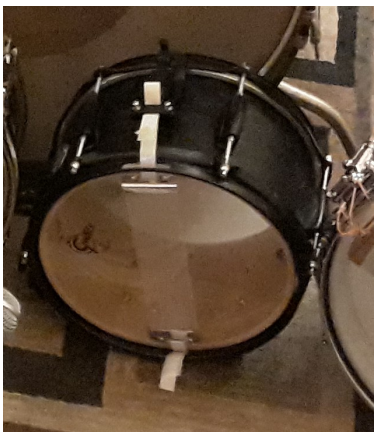
INSTRUMENT BANK

The instrument bank includes 31 instruments. 01-Frankensnare is a multilayered instruments allowing up to three snare drums to be layered, plus claps and guitar chiks. 02 through 30 are individual drums loaded one at a time – the same samples which can be selected for Frankensnare layers, but with more detailed control. The final instrument, 99-Extras, is only the claps and guitar chiks.



DRUM LIST

- 01-Frankensnare is the layering instrument, explained in more detail in the Controls section. All of the drums listed below can be selected as layers within it.
- 02-10x6 ash snare brushed is a 10” diameter, 6” deep Gretsch ash snare played with a brush. Only hits are sampled, no stirs – the very small size of the drum makes performing stirs impractical. Six velocity layers for the main hits, and five for the sidesticks. Four round robins for all snares.



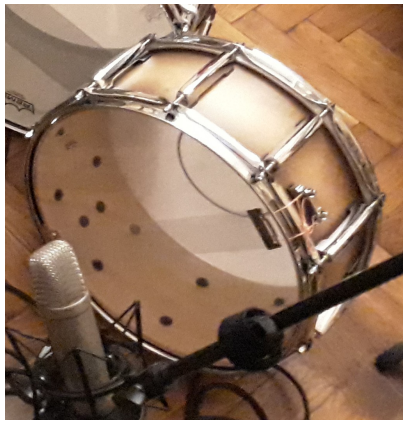
- 03-10x6 ash snare is the same snare as above, but played with a stick. Small but fairly deep, this is a „popcorn” snare with a short, bright sound. Most drummers would use such a snare as a „side” snare and not the main snare in a kit. Six velocity layers for the main hits, and five for the sidesticks.
- 04-12x5 steel snare is a 12” by 5” Pearl steel snare. Again, it has a short, bright sound and would normally be used as a side snare, though its owner has used it as the only snare on many small gigs. Seven velocity layers for the main hits, and four for the sidesticks.



- 05-14x5 maple snare brushed is a 14” by 5” DW maple snare from a Frequent Flyer kit, played with a brush. This time both hits and stirs are available. Five velocity layers for the main hits and for the sidesticks.



- 06-14x5 maple snare is the above snare, played with a stick. It's a fairly shallow but full-size snare, with a sound that's a little on the bright side. Seven velocity layers for the main hits, and five for the sidesticks.
- 07-14x6.5 maple snare is a 14” by 6.5” Pearl Masters maple snare. This is perhaps the most middle-of-the-road of all the snares here. Eleven velocity layers for the main hits and six for the sidesticks.



- 08-14x6.5 maple snare muted is the above snare, muted heavily with a kitchen towel, for a shorter and darker sound. Seven velocity layers for the main hits, and six for the sidesticks.

- 09-14x8 aluminum snare is a 14” by 8” Pearl Free Floating aluminum snare. It has a short, very controlled and fairly deep sound. It also has more snare wires on the bottom than any of the other snares sampled here, so even lighter hits produce a good amount of snare noise. Eleven velocity layers for the main hits, and five for the sidesticks.



- 10-14x8 aluminum snare rimshot is the same snare as above, only with rimshots instead of regular hits. Three velocity layers for the rimshots, and five for the sidesticks.
- 11-14x8 birch snare is a 14” by 8” Szpaderski birch snare with the top head tuned low. It has a very warm sound with a fair amount of thump, especially in this low tuning. The internal mute was used to keep the sound short. Nine velocity layers for the main hits, and five for the sidesticks.



- 12-18x16 prepared birch tom is an 18” by 16” Szpaderski birch floor tom which sounds somewhat snare-like because we placed various objects on the top head. There are the original broken snare wires from the above snare, the snare block from a cajon placed wires side down, two brushes, and several wrenches. It makes a deep, short and noisy thud. No sidesticks were sampled here, because there wasn't enough room left on the head. There are seven velocity layers.



- 13-20x6 poplar snare brushed is a 20” by 6” poplar snare drum, made by Bang! Drums Burgdorf by cutting down a kick drum shell and using it to build a huge snare drum. This is played with brushes, and includes both hits and stirs. We put an imitation calfskin Evans head on the top, for stir-sampling purposes. The main hits have seven velocity layers, and the sidesticks have six.



- 14-20x6 poplar snare is the above snare played with a stick. It has a very deep sound with a lot of low end. The main hits and sidesticks both have six velocity layers.

- 15-20x6 prepared poplar snare is the above snare, partly disassembled and with the snare wires laid loosely across the top head, and the snare block from a cajon laying on the other side of the head. The bottom head was removed, and the drum was laying on a carpeted floor to kill the resonance. All this results in a very short, thumpy sound vaguely reminescent of an 808 drum machine snare. The sidesticks here are different sidestick samples than used for the above two instruments. The main hits have six velocity layers, and the sidesticks have four.



- 16-80s sympathy snare is actually a 20” by 20” DW kick drum, surrounded by several snare drums. The concept is to make sympathetic snare buzz the main feature of the sound instead of just an incidental noise – hence the large number of snare drums, which were also tuned to maximize buzz with this particular kick. The resulting sound is reminescent of a synthesized snare with a noise oscillator imitating the snare wires. The snares are: a Ludwig pre-serial keystone snare from approximately 1963, a Taye 14” by 4.5” snare, an unknown homemade snare drums dating likely to the 1970s, an Apollo 13” by 6” snare, a DDrum 13” by 5” snare, a DW 13” by 6” snare drum from the late 1980s, and a Remo PTS 16” by 16” „snom” - a floor tom modified by adding snare wires. No sidesticks, for obvious reasons. Seven velocity layers.



- 17-60s sympathy snare is a 22” by 16” Apollo (Japanese stencil kit) kick drum surrounded by the above snare drums, though in somewhat different positions, and retuned. The sound is similar to the 80s version, but with more depth and longer sustain. Three velocity layers.

- 18-13x9 birch snare. This fat pink little piglet was converted from an Latvian RMIF tom by adding some parts from a cheap DiMavery snare. Sounds short and thumpy. A drummer friend described the sound of the bottom mic as „pixelated”. Six velocity layers for the main hits, five for sidesticks.



- 19-13x9 birch snare rimshots. Rimshots played on the above snare. Six velocity layers.
- 20-13x9 birch snare with tambourine. Same snare again, with a tambourine laying on top of it. A dirtier sound than layering a snare with tambourine samples. Seven velocity layers.
- 21-14x5 mystery wood snare. A very unusual snare from the Ton factory in Poland, made in the 1950s or 1960s. The rims are massive, there are only five lugs per side, the snare wires are inside, and there's a big internal muffler right in the center of the head. It was tuned very low, because we didn't want to risk damaging the original skin heads, which don't even appear to be calfskin – perhaps goat. Due to the low tuning and thick heads, it sounds quite thuddy. The sound is downright faux-medieval, even though this snare was an attempt at the most modern innovation during the atomic age. Six velocity layers for the main hits, four for sidesticks.



- 22-16x7 sapele snare. Another snare we made from parts – the decades-old shell was refinished and lugs installed by Fat Flying Drums, and we added the rest of the hardware. Oversized but tuned quite high here, for a surprisingly balanced and middle-of-the-road sound. We liked it so much it ended up with eleven velocity layers for the main hits, and seven for the sidesticks.



- 23-20x12 alder snare. This started out as a Polmuz bass drum. A previous owner added an extra set of lugs to one side, enabling it to be tuned extremely high, and we added metal hoops, an internal baffle, and two sets of microsnare wires. Thumpy and short, again reminescent of an 808 drum machine snare. Seven velocity layers for main hits, five for sidesticks.



- 24-20x12 alder snare, tight spring. The above drum, hit directly on top of one of the sets of microsnare wires which is installed so it's pushed very tightly against the head. A strange, metallic sound. Six velocity layers.
- 25-22x5 mahogany snare, brushed. Another strange beast we assembled using a shell converted from a kick by Fat Flying Drums, a very unusual triple-flange hoop from a RMIF kick drum on the bottom, a wood hoop on the top, and two sets of snare wires tied together in the middle. Huge and dirty. We used the same drum in our Unruly Drums kit, but here it is tuned much higher and sounds quite different. Seven velocity layers for main hits, five for sidesticks.



- 26-22x5 mahogany snare. The above drum, dampened with a cloth and hit with a stick. The tuning is so high the top head even gets a little ping though the drum's size obviously gives it a big bottom as well. Eight velocity layers.
- 27-22x5 prepared mahogany snare. The same snare again, but muted by laying a wide set of 15” snare wires on the top head, rather than a cloth. This keeps the top from having an audible ping, adds more buzz. The end result is probably the closest to a giant version of a normal snare sound in all of Frankensnare. We also recorded sidesticks with this configuration. Seven velocity layers for main hits, five for sidesticks.
- 28-22x14 prepared birch kick.



Before we cannibalized that RMIF kick for its hardware, we also set it up on its side, poured 400 grams of rice and several hundred paperclips inside, put the batter head back on, and tried to use all that as a snare. It worked as long as it was hit gently. The sound is very unusual – a lot of thud from the big drum, and a rather chaotic buzz that gets looser and more scattered with the harder hits. No sidesticks. Eight velocity layers.

- 29-ziggurat, level one.



This is a big beast built of three layers – the 22x5 snare with the 16x7 snare on top of it, and an approximately 14” folk drum made by one of our ancestors in the 1940s from a sieve hoop as the top layer. Yes, really. This was recorded by hitting the area of the 22” snare that's sticking out from under the 16” snare. It's a surprisingly bright, snappy sound. No sidesticks. Eight velocity layers.

- 30-ziggurat, level two. The same configuration as above, but recorded by hitting the low-tuned canvas head of the top drum, for a thuddier sound with less snare buzz. Eight velocity layers.
- 99-extras contains claps, guitar chiks and tambourine. There are no velocity layers here except for the tambourine, which has five velocity layers and five round robins. The first layer of claps has five round robins, the second has four, and the third has three. The stack claps (a stack of cheap cymbals which sounds vaguely like an early digital clap) and chiks also have five round robins.

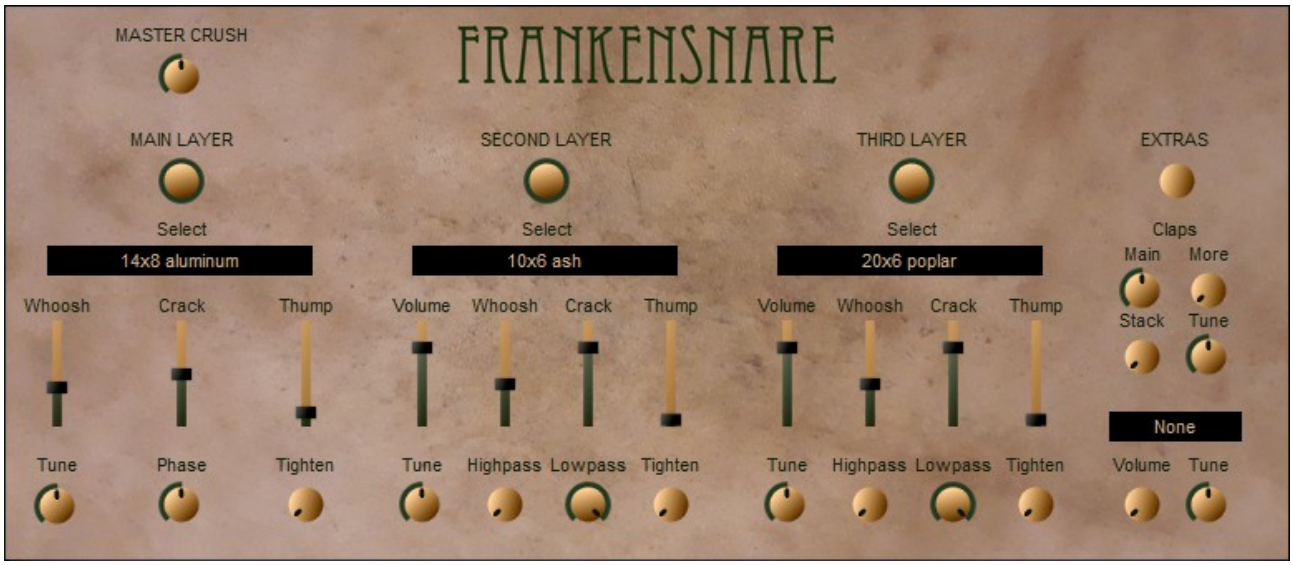
NOTE MAPPING

Most of the drums use only two notes – MIDI note 38 is a regular snare hit, and MIDI note 37 is a sidestick. This allows General MIDI compatibility. A few instruments do not have the sidestick. In the Frankensnare instrument, only the main layer is used by the sidestick note, so sidesticks are not layered.

In addition, two of the individual brushed drums also include stirs, which use MIDI notes 24 and 28 to start stirs, and MIDI note 26 to mute them. MIDI note 24 is a long, slow stir and MIDI note 28 a faster, louder stir. One stir can be played at a time – playing another stir will mute the currently playing stir. MIDI note 26 mutes the current stir without making any additional sound.

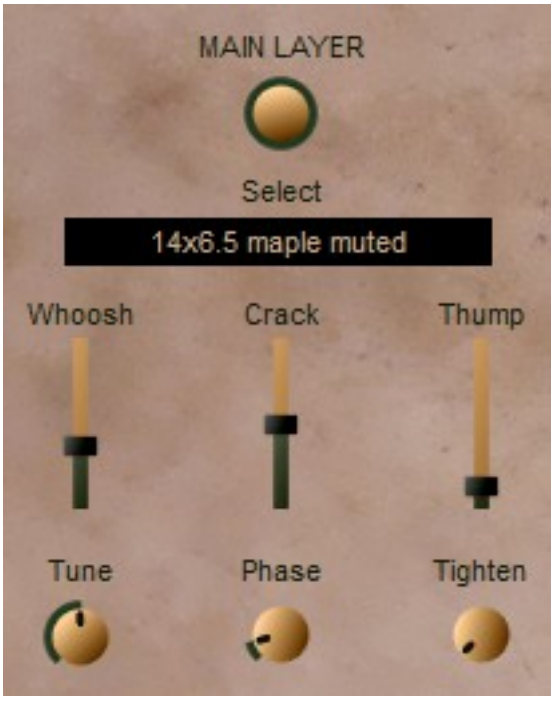
Controls

Frankensnare

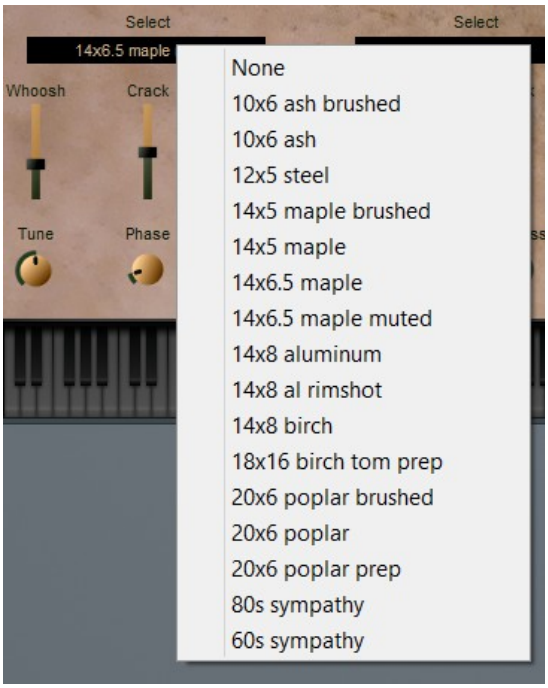


The Frankensnare has a large number of controls, with the other instruments being somewhat simpler.

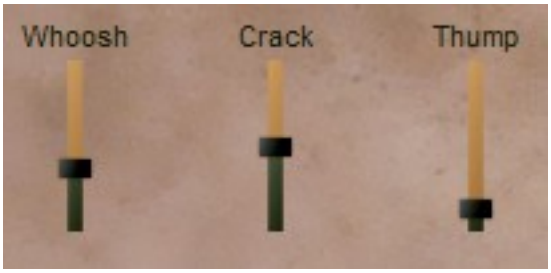
MAIN LAYER



The controls for the main layer start with an on/off button at the top – these can be used to quickly mute some of the layers to isolate the sound of a particular layer, or to create instruments which don't use all the layers.



Below that is a selection box which is used to select which set of drum samples is used for the main layer. The instruments are listed in the drum list above.



Next are three controls: Whoosh, Crack and Thump. These are mostly volume controls, and their precise effect depends on which drum is sampled. For all the layers, Whoosh is „mostly" overhead mic volume, Crack is snare bottom mic volume, and Thump is snare top mic volume. However, that's just „mostly". What they do in the main layer is more precisely described below.

In general, Whoosh up to 50% turns up the volume of the overhead mics. Between 50% and 75% it adds another layer of the overhead mics transposed an octave down – a huge, reverberating sound. Above 75% usually it adds the snare bottom mic transposed an octave down – a dirty, industrial sound that with some drums resembles a huge sheet of metal being hit.

With the 18x16 prepared tom, the top mic is used for Whoosh instead, and with the 20x6 prepared snare, the closest mic is used as there is no bottom. With the sympathy snares, the overheads and snare clusters mics are used up to 75%, and both the main snare bottom and snom bottom mics are used above 75%. The 60s sympathy snare also brings in the room mic here, with no transposition.

Crack up to 50% usually turns up the volume of the snare bottom mic, and above 50% it adds the snare bottom mic transposed an octave up. With the 18x16 prepared tom and 20x12 snare, the top mic is used here instead, and with the 20x6 prepared snare, the closest mic is used as there is no bottom. With the symphathy snares, the snare bottom and snom bottom mics are used here.

Thump is usually the snare top mic. With the 18x16 prepared tom and 20x12 snare, the bottom mic is used here instead, and with the 20x6 prepared snare, the close mic is used as there is no bottom. With the sympathy snares, this control brings in the kick in mic up to 50%, and the kick out mic above 50%, so with the thump turned up there is a full-blown kick drum sound.

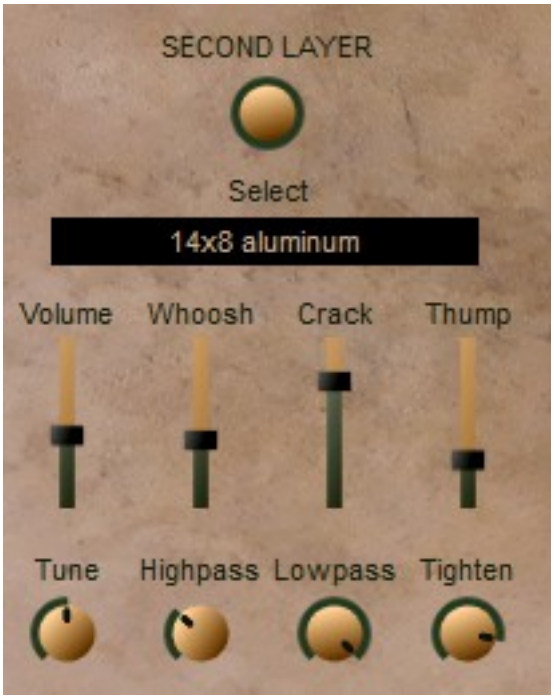


The Tune control is simple – a tuning control from one octave down to one octave up.

The Phase control aligns the phase of the overhead mics with the top and bottom close mics. At 50%, the alignment is realistically as recorded. Below 50%, the overheads are delayed slightly, and above 50% the overheads are brought forward in time, until at 100% they are perfectly aligned with the close mics. This, basically, emulates how microphone signals would be aligned if the speed of sound was infinite. The top and bottom mics (or close and closest mics, for the 20x6 prepared snare) were already aligned during the sample editing. With the sympathy snares there are a lot more mics to align, so each mic is shifted by a different amount. The more in-phase things are, the more high-fidelity and clean they sound.

The Tighten control applies a volume envelope to the layer, to emulate muted drums. At settings close to 100%, only a very short transient remains.

SECOND AND THIRD LAYERS



The second and third layers have identical controls. There's an on/off button at the top and a selection box which work the same as they do for the main layer. The volume control is a volume for the entire layer.

The Whoosh control is simply a volume control for the overhead mics (or overheads plus snare cluster mics for the sympathy snares) with no transposed copies brought in at higher levels. The 60s sympathy snare also includes the room mic under this control, and both sympathy snares include the snare cluster mics.

The Crack control is a volume control mainly for the bottom mic, with no transposed copies. For the 18x16 prepared tom, it controls the top mic. For the 20x6 prepared snare, it controls the closest mic. For the sympathy snares, this controls the snare bottom and snom bottom. So, in general, the Whoosh and Crack for the second and third layers aren't quite as whooshy or cracky as they are for the main layer.

The Thump control is a volume control mainly for the top mic. For the 18x16 prepared tom, it controls the bottom mic. For the 20x6 prepared snare, it controls the close mic. For the sympathy snares, this controls the kick in mic up to 50%, and blends in the kick out mic above 50%.

The Tune and Tighten controls work the same as they do for the main layer. The Highpass and Lowpass controls filter out low and high frequencies, respectively, so that only a specific frequency range from the layer can be used.

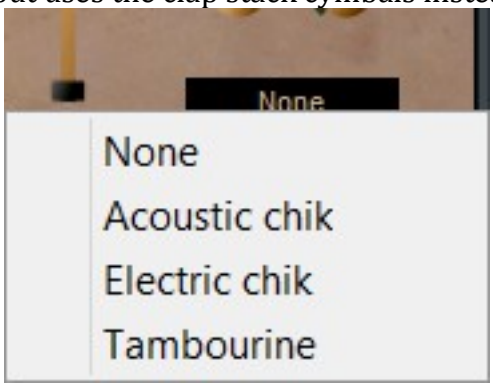
Though they have similar controls, there is one difference between the second and third layer. The first and third layer use four round robins for all snares, and the second uses three. This means the layers' round robins don't align with each other, and the resulting sound has 12 virtual round robins before an identical hit repeats.

EXTRAS LAYER



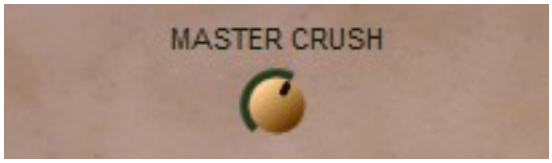
The extras layer has an on/off button at the top. Below that are two volume controls for the claps. The Main control brings in the basic claps up to 50%, and the same claps transposed an octave down above 50%. The transposed claps, like the transposed overheads of the main layer, result in a large, reverberating sound.

The More control brings in a second set of claps up to 50%, and a third set above 50%. The Tune control works the same as the Tune controls for the first three layers. The Stack control works like the Main control, but uses the clap stack cymbals instead of „real" claps.



Below the claps, there's a selection box for one more layer. Currently there are three choices: muted chiks recorded on either acoustic guitar or electric guitar, and tambourine. The selected sound also has volume and tune controls.

MASTER CRUSH



The Master Crush knob in the Frankensnare is a gain control for a limiter effect. At 50% it has no effect on the sound, and below 50% it reduces the volume. Above 50% is where it boosts the signal, and the limiter kicks in.

SINGLE SNARES

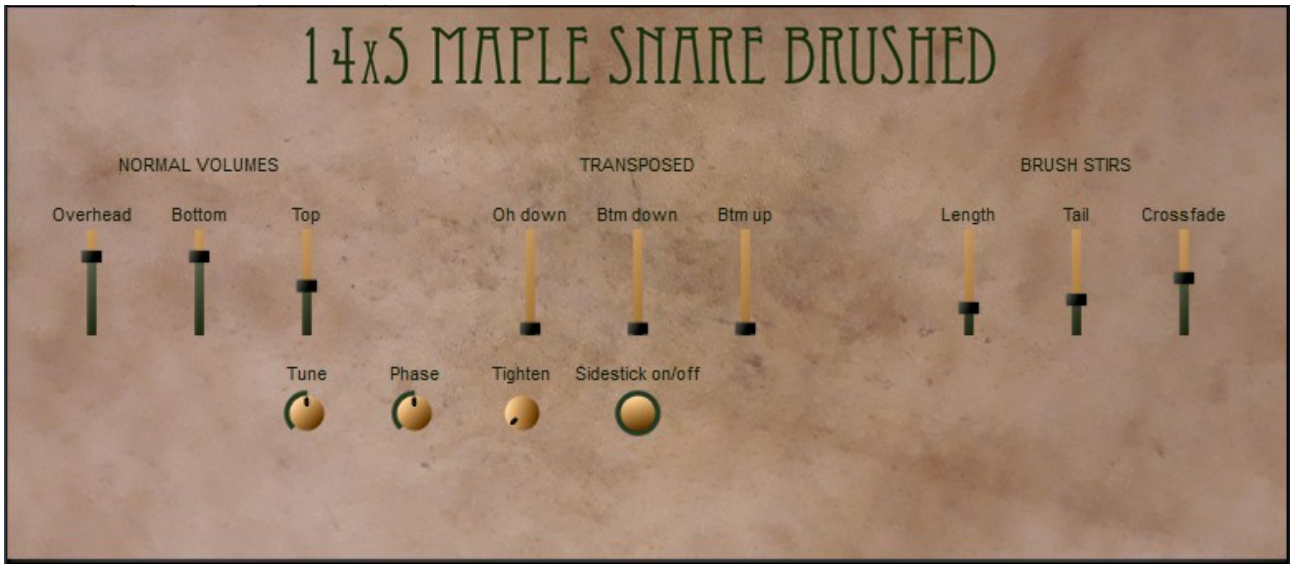
BASIC CONTROLS



The single snare instruments have generally similar controls, with a separate volume control for each mic (or stereo pair of mics), and additional volume controls for the transposed mics where that makes sense. The sympathy snares have a very large number of mic controls, as they were recorded with a large quantity of mics.

The Tune, Phase and Tighten controls work in the same ways the do for the Frankensnare main layer. The Sidestick on/off button enables the sidestick to be switched off. This can be convenient when sending the same MIDI data to multiple instances of Sforzando, so that only one instance will actually produce sound when a sidestick note is sent. Although sidesticks can be layered, we generally feel it's not a good idea most of the time. The 18x16 prepared tom, 20x12 snare, sympathy snares and ziggurat don't have this button, as they have no sidestick samples.

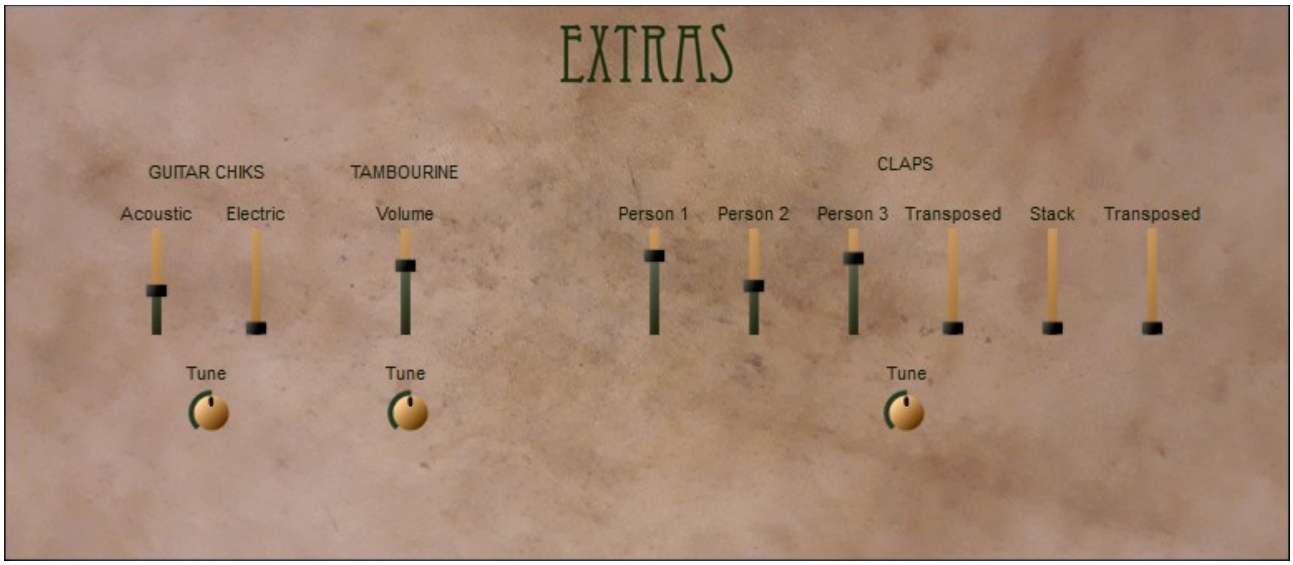
BRUSHED STIRS



The 14x5 and 20x6 brushed snares have additional stir controls, which are used with MIDI notes 24, 26 and 28 to play stirs – performed in the real world by making circles on the head of the snare with a brush that's in constant contact with the head. The Length control sets how long it takes for the stir to reach peak volume, the Tail sets how long the volume takes to decay after the peak, and the Crossfade control sets how quickly an existing stir fades when muted.

The Length, Tail and Crossfade controls have no effect on the regular hits and sidesticks, and likewise the Tighten and (obviously) Sidestick on/off have no effect on the stirs.

EXTRAS



The 99-Extras instrument has a separate volume control for each flavor of guitar chik, tambourine, each set of claps, and for the claps transposed an octave down. There's also one Tune control for the chiks, one for the tambourine, and one for the claps.

SAMPLER FEATURES

The below are standard features of Sforzando, and are explained in more detail in the Sforzando manual. This is just a short summary of how these features apply to Secret Agent Guitar.

SNAPSHOTS

Specific control settings can be saved using Sforzando's Snapshot function – basically, these are presets. This saves a snapshot of the instrument currently loaded into one instance of Sforzando and its MIDI CC settings. Several factory snapshots are also included.

EFFECTS AND SETTINGS TABS

The Effects tab and Settings tab contain default Sforzando functions. Effects can be used to add more reverb (together with the Send control at the top of the Sforzando interface). Settings can be used for monitoring adjusting RAM usage and polyphony, though in general, the default settings appear to work well when using one guitar in a project.

Increasing the maximum engine RAM may be needed in projects which use multiple instances of Sforzando and other ARIA engine products which combine to use a large amount of RAM. Increasing polyphony can be useful when playing rolls, especially with the sympathy snares, but is probably not necessary.

CREDITS

Sympathy snares recorded by Robert Randolph (https://www.admiralbumblebee.com). The rest recorded by Drogomir Smolken, Paul Chapman and Adam Sobolewski. Presets by Drogomir Smolken and Triller (https://soundcloud.com/trillergs). Thanks to all our beta testers, to Hannes of Bang! Drums Burgdorf (https://twitter.com/bangdrumsbdorf) for loaning us the giant 20" snare, and to Grzegorz Kurpiel of Fat Flying Drums (https://pl-pl.facebook.com/fatflyingdrums) for making custom shells for us.

APPENDIX

FRANKENSNARE MIDI CC ASSIGNMENTS

7	Volume
10	Pan
51	Main On/Off
52	Second On/Off
53	Third On/Off
54	Extras On/Off
71	Extra Select
72	Extra Vol
74	Extra Tune
75	Clap Main
76	Clap More
77	Clap Stack
79	Clap Tune
80	3 Select
81	3 Whoosh
82	3 Crack
83	3 Thump
85	3 Tune
86	3 Vol
87	3 HP
88	3 LP
89	3 Tighten
90	2 Select
92	2 Crack
93	2 Thump
94	2 Whoosh
95	2 Tune
96	2 Vol
97	2 HP
98	2 LP
99	2 Tighten
100	Main Select
101	Main Whoosh
102	Main Crack
103	Main Thump
105	Main Tune
107	Main Phase
109	Main Tighten
400	Limiter Threshold
401	Limiter Level

Individual Snare MIDI CC Assignments

7	Volume
10	Pan
20	Stir Length (14x5 and 20x6 brushed snares only)
21	Stir Tail (14x5 and 20x6 brushed snares only)
22	Stir Crossfade (14x5 and 20x6 brushed snares only)
101	Overhead Vol
102	Bottom Vol
103	Top Vol
105	Tuning
107	Phase
109	Tighten
111	Sidestick On/Off (except for 18x16 prepared tom)
112	Overhead Low Vol
113	Bottom Low Vol
114	Bottom High Vol

Sympathy Snare MIDI CC Assignments

7	Volume
10	Pan
101	Overhead Vol
102	Bottom Vol
103	Kick In Vol
105	Tuning
107	Phase
109	Tighten
110	Kick Out Vol
111	Snom Vol
112	Cluster Vol
113	Room Vol (60s sympathy snare only)
114	Overhead Low Vol
115	Bottom Low Vol
116	Snom Low Vol
117	Cluster Low Vol
118	Bottom High Vol
119	Snom High Vol
120	Cluster High Vol

EXTRAS MIDI CC ASSIGNMENTS

7	Volume
10	Pan
71	Acoustic Chik Vol
72	Electric Chik Vol
74	Chik Tune
75	Clap One
76	Clap Two
77	Clap One Low
78	Clap Three
79	Clap Tune
80	Clap Stack
81	Clap Stack Low
82	Tambourine
83	Tambourine Tune